

Composing translated partial swaps with semantic precision and bulk resampling

Zhihao Chen (jacklightChen)
With substantial OpenAI Codex assistance

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Abstract

The unchanged finite bit and complex interfaces of PR21 compose with the semantic precision, arbitrary-coordinate routing and bulk resampling interfaces exposed by the RaD contribution in PR20. We supply the circuit-specific scalar charge, the larger-child precision induction, the simultaneous nested row stock and a strict rational assembly. The resulting conditional exponent saving is $\kappa = 1099/10^8 = 1.099 \cdot 10^{-5} > 2^{-17}$. The original FFT prefix layout suffices. This is a composition of attributed constructions with written proofs and executable checks, not formal verification or completed external mathematical peer review.

1 Statement, provenance and proof boundary

Theorem 1. *Under the retained multiplication framework and its fixed finite-alphabet multitape interfaces, with the attributed phase-cell inverse, routing and bulk resampling arguments identified below, integer multiplication has*

$$T(n) = O(n(\log n)^{1-\kappa}), \quad \boxed{\kappa = \frac{1099}{10^8} > 2^{-17}}.$$

All finite graphs, bases, native tables and tape counts are fixed independently of input length. Their constructive setup and the original analytic and exact-recovery thresholds remain eventual conditions.

The finite bit and phase networks are precisely PR21, commit `5ba6cf0bfb68f2be8d15610e7972207c50254d6a`.

Their complete proof is retained in `notes/translated-partial-note.tex` and its compiled PDF; no gate, physical edge, basis, source frame, scalar field or phase sign is changed. PR18's partial swaps and PR10's batching are due to icekylinx; the source frame idea is due to eumemic's PR13; the translated endpoints and complex phase construction are Zhihao Chen's PR21, following PR7 and PR16. Rohan Arun's PR19 independently transfers partial swaps to the ternary network. Its stated $2.093495 \cdot 10^{-6}$ saving is not an additional independent saving to add to PR21, and its scalar producer is not used here.

The new analytic dependencies come from the RaD project (hipotures), exposed by PR20 and pinned at RaD commit `45d9b60355872041f9275f77c057514def0d45bc`. Its recorded Codex and GPT-6.1-Sol Ultra assistance remains attributed. Unmodified arguments, source code, CC0 license and a hash manifest are included in `references/rad20`. The source package's historical status strings describe its own research sequence. We use its completed reviewed constructions, and

explicitly re-establish the parameters that change for the PR21 networks. The term “review” in those filenames refers to their recorded computational/agent review, not external human review.

Our new contribution is this compatibility argument, exact scalar and row accounting, and assembled witness. The older GPT-6 Astra attribution is preserved as recorded, without identifying the current runtime model. Future work using this contribution should acknowledge Zhihao Chen (jacklightChen) and all other dependencies used. This request adds no license condition and asserts no worldwide priority.

2 Unchanged finite interfaces

Let $a = 11/10^6$ and $b = 18/10^6$, with $\tau = 1 - a$, $\sigma = 1 - b$. PR21 provides arbitrary-width equal-chunk interchange with cost $O(Vu^\tau)$, including arbitrary complete spectator gaps, padding and scratch restoration. Its exact bit characteristic is strictly below one. Its parameters are

$$m_b = 32775, \quad W_b = 5820554077800, \quad r_b = 32729.$$

The retained common rational basis supplies actual contiguous recursive blocks. No edge-specific basis or uncharged gathering is added.

The independent complex network has

$$m_c = 21952, \quad W_c = 2085111546336, \quad s_c = 45772350635112192, \quad r_c = 21924.$$

Its complete physical child histogram has weighted rank sum s_c and strict homogeneous characteristic at exponent σ . All its physical edges, both orientations, shared-bank boundaries and endpoint phases were checked for PR21. Its source-translated stage-two auxiliary still has physical endpoint C_I , on arbitrary dirty Gaussian-dyadic streams. Thus each completed recursive call is exactly the tensor phase operation on its selected axes and the identity on its other coordinates. The signed residual implementation is $(-i)^{f^n} Z_- C^{\otimes rf} Z_-$ with the paid binary adapters. It never substitutes a characteristic-zero address projection for a binary phase operator. Every recursive child has $rf < e$ selected positions.

The rational and binary payload/address interfaces keep their distinct fields. The RaD router consumes the arbitrary-width bit interchange contract, independently of its internal payload alphabet. The semantic precision argument consumes the exact completed complex contract, independently of how its finite residuals were constructed. These are the interfaces at which the two packages compose; their graph budgets are not mixed.

3 Linear semantic precision for this actual circuit

Every completed forward or inverse child on u selected axes has Gaussian coefficients in $2^{-u}\mathbb{Z}[i]$ and absolute row sum at most 2^u . Its unit sign wrappers do not increase that bound. With incoming magnitude 2^t and denominator exponent $p + t$, a returned stream therefore has magnitude at most 2^{t+u} and denominator exponent at most $p + t + u$. Internal intermediate values may be larger. They are never returned as unfinished operands to the parent.

All computation uses the same fixed fine grid, with exact internal halves. The argument changes a bound on a returned value, not the stored encoding. No child is rounded, compressed or re-encoded at its return. Other parent operands remain parked on the fixed stack tapes until the child completes. The scalar schedule restores arbitrary dirty auxiliaries, so this statement holds on every physical role, rather than on a selected zero-scratch output.

Here the local PairedComplex28 circuit has 64298 active nodes, 3276 data triples, and 93838 side roles. Each of its three tensor stages uses four mixer sweeps, two copies, two injections and four grouped central gates. Counting input nodes as mixer gates only increases the bound. Consequently

$$G = 3 \binom{28}{3}^2 (4 \cdot 64298 + 4 \cdot 3276 + 4)$$

bounds its grouped scalar gates. Put

$$E = 64(W_c + m_c + 1)^3, \quad B = s_c + E, \quad C_0 = 32m_c B^2.$$

Exact integer arithmetic verifies

$$2GW_c^2 + 8s_c + 4W_c + 4 + 32m_c < E.$$

This conservatively pays scalar evaluation, all unit phase wrappers, role corrections, and the fewer than m_c tail kernels. In particular a smaller physical register count is not substituted for a gate count.

Let $A(e)$ bound temporary magnitude and denominator excess for an e -axis call. A leaf has $A(e) \leq 8e$. Before any active child, the total selected widths of already completed calls are at most $s_c \lfloor e/m_c \rfloor$. The active call has at most $r_c \lfloor e/m_c \rfloor$ axes. The correct recurrence is therefore

$$A(e) \leq A(r_c \lfloor e/m_c \rfloor) + s_c \lfloor e/m_c \rfloor + E.$$

Writing $f = \lfloor e/m_c \rfloor \geq 1$, induction gives $A(e) \leq 2Be$, since

$$2B(m_c - r_c) \geq s_c + E, \quad 2Br_c f + s_c f + E \leq 2Bm_c f \leq 2Be.$$

This argument explicitly accommodates a child almost as wide as its parent. It holds for any fixed stopping exponent. Completed disjoint selected-axis pieces have total width at most d ; prefix kernels, outer phases and final normalization contribute the retained $18d$. Hence

$$A_{\text{layer}} \leq (2B + 18)d < C_0 d, \quad C_1 = 1.$$

The condition $\epsilon < 1$ makes this guard eventually $O(p)$ for $d = \lfloor b_{\text{in}}^\epsilon \rfloor$, $p = 6b_{\text{in}}$. The old PR21 path guard remains a valid older estimate but cannot be used with the new near-one choice of ϵ .

4 Complete simultaneous rows and the stopped tree

Exact integer powers establish the least halving depths

$$m_b^{494} > 2r_b^{494}, \quad m_c^{544} > 2r_c^{544}.$$

Also $W_b < 2^{43}$ and $W_c < 2^{41}$. At a top-level call, reserve the product $Q = W_c^{D_c} W_b^{D_b}$, with $D_c = 544 \lceil \log_2 e \rceil$ and $D_b = 494 \lceil \log_2 e \rceil$, using a conservative global width bound. The bit factor is consumed temporarily by an adapter and restored before another complex child proceeds. Using the maximum of the two stocks would not establish this nested contract.

There is a physically preceding row prefix: apply exact individual C kernels once to the first $q_0 = \lceil \log_2 Q \rceil$ selected positions, retaining their full chunks. If fewer positions exist, this is the whole $O(V \log p)$ elementary fallback. Otherwise these transformed chunks supply a leading row field of cardinality $R \geq Q$. Pad complete rows once to $Q \lceil R/Q \rceil < 2R$. Descendants inherit the

same stock, with complete active, control and spectator fields in every role. Their exact all-role return restores appended zero rows to zero, so they can be removed at the completed boundary. The prefix is neither moved from a suffix for free nor transformed again inside descendants.

For eventual $e \leq Cp$, $p \geq C$, $\log_2 p \geq 25$,

$$\log_2 Q \leq (43 \cdot 494 + 41 \cdot 544) \frac{51}{25} \log_2 p < 89000 \log_2 p.$$

Thus the safe combined stock is p^{89000} ; the supplementary suffix condition is $b_{\text{in}}^{1-\epsilon} > 356000(\log_2 b_{\text{in}} + 8)$. Its compressed integer certificate is supplied, rather than borrowing PR20's stock for a different graph.

For the stopped complex tree, give each node potential equal to its relative logical volume times e^σ . The strict homogeneous characteristic contracts the sum of child potentials. Complete leaves at $e < d^\beta$ have total cost exponent $\sigma + \beta(1 - \sigma)$. Since $\tau > \sigma$, the paid bit-adapter cost sums geometrically with exponent τ . Compact reservation costs have exponent $\max(1 - c, 0)$. These conclusions permit unequal child depths, retain every binary adapter and tail scan, and yield the three exponents used in the final certificate.

5 Attributed movement, inverse and bulk interfaces

The following are the RaD analytic and tape constructions, not new finite-graph claims. Their full written arguments and discriminating controls are pinned in [references/rad20](#).

Coordinate routing. The active-masked compact gadget implements disjoint-source XORs, with arbitrary dirty reservoirs. A completed four-update identity restores guards and temporary digits, so a source in an inactive slot or guard is read consistently by the load and unload. Actual modular rotations are bijections. Correct bad addresses by TS^{-1} evaluated at the current address, using stable radix reinsertion into the bad holes. The density bound is local to each invocation; every extraction and reinsertion scan is paid. Three untouched reservoirs partition a matching, and two matchings implement a coordinate permutation. For short coefficient records, a separate untouched spectator field of width $\ell = \Theta(p^{1-\epsilon})$ enlarges records without enlarging volume. Original chunk swaps place and restore that spectator; the proof is not circular. The small-width fallback requires $\epsilon > a$. The resulting cost is $O(Vp^\tau \text{polylog } p + \text{poly } p)$ for known coordinate permutations on complete rectangles.

Phase-cell inverse. RaD's rounded structured global matrix H has exact diagonal one, row gap at least $1/(8p)$ and inverse row norm at most $8p$. Its phase cells preserve the physical period cut. Truncated Toeplitz interiors, Schur complements and rounded banded boundary solves operate with $O(p)$ -bit online coefficients; exact rational setup is reused once per axis and is $n^{o(1)}$. The rounded similarity weights are downward approximations with a four-ulp allowance, not assumed exact irrational roundings. With $\alpha^2 = \Theta(p^r)$, the band satisfies $w^2 = O(p^{1-r})$. The normalized inverse has norm below $1/8$; the logarithmic normalization term is retained in $\gamma = d(2\alpha^2 + \lceil \log_2(32p) \rceil + 1)$. Thus $\epsilon + r < 1$ is essential.

Bulk locality and tape movement. Take target core length $L = 2^{\lceil 8 \log_2 p \rceil}$ and halo $A = 4096p^3$. Every principal window restricts the same global rounded H . With $J = 512p^2$ and artificial-boundary distance at least Jw , the two Neumann tails give row error at most $32p 2^{-J/(16p)} < 2^{-16p}$. The

complementary period is greater than w , and the original phase cut is never merged after lifting a wrapped window. Local normalized maps retain norm headroom; tensor errors add in row norm.

Near-one source padding fills each source halo into exactly L entries, using $L \geq 64dA$. Target halos instead have persistent length $L + 2A$; their total tensor volume is $T(1 + 2A/L)^{d-1} < 2T$. Only the currently moved short field is temporarily padded to binary length. Simultaneous binary padding of all halo fields is invalid and explicitly rejected. Two alternating overlap tapes emit monotone windows without shifting a FIFO or rereading an entire line. Factor pages are scanned sequentially in their original coarse-axis order and reused over complete spectator fields. An axis is cropped only after its own inverse completes; remaining axes retain their halos. The final ordered cores restore the original prime-box row order and normalization.

A constant number of full coordinate routes and the short-field passes therefore give the complete bound

$$O\left(Tp[p^\tau \text{polylog } p + d \text{polylog } p + dp^\delta + w^2p^\delta + dw^2p^\delta/L]\right) + n^{o(1)}.$$

All the displayed terms occur in our assembly. The phase-cell proof uses an unconditional conventional multiplier for its small convolutions, so there is no circular invocation of the new multiplication theorem.

Prime supply. The old restriction $1 - 2\epsilon > 0$ is replaced, not silently dropped. Baker–Harman–Pintz, Theorem 1, supplies a prime in $[x - x^{21/40}, x]$ for sufficiently large x . Place d disjoint such intervals in the required window of length $x/(4d)$. The sufficient inequality $12d^2 < x^{19/40}$ holds eventually when $x = 2^{\Theta(p^{1-\epsilon})}$ and $\epsilon < 1$. The required distinct odd primes, near-one ratios and deterministic $n^{o(1)}$ search remain available. Their theorem’s eventual threshold is retained separately from our explicit power tests.

6 Exact strict assembly

Choose

$$\begin{aligned} h &= 10^{-8}, \quad \beta = \frac{1}{4}, \quad q = a(1 - 2h), \quad \lambda' = 1 - q, \quad \lambda = \frac{1}{2}(\tau + \lambda'), \\ c &= q(1 + h), \quad \epsilon = \frac{1 - h}{1 + c + q}, \quad G = \epsilon q, \quad r = \frac{G + 1 - \epsilon}{2}, \quad \delta = h/8. \end{aligned}$$

Here $\epsilon \simeq 0.9999779904845393$ and $r \simeq 1.6504636568 \cdot 10^{-5}$. The retained original prefix is sufficient; no changed balanced-layout estimate is needed. The seven margins are

operation	saving
original prefix	$1 - \epsilon(1 + c)$
coordinate movement	a
compact phase layer	ϵq
bulk exposure	a
Gaussian arithmetic	$\min(1 - \epsilon - \delta, r - \delta)$
scalar work	$1 - \epsilon - \delta$
dimension	ϵ

The exact minimum is G , with

$$G - \frac{1099}{10^8} > 9 \cdot 10^{-9} > 0.$$

The program recomputes 47 positive constraints, including these seven strict absorption gaps, all exponent orderings, $(1 - \beta)b > a$, $c > q$, the linear guard, compact geometry, cell/band separation,

short-record fallback, artificial microbox boundaries and row-stock allowance. These replace the old separate-exposure/old guard constraints using the written constructions above. Negative controls show that the old guard, old separate exposures, and claimed 2^{-16} saving each fail. The strict gaps absorb fixed logarithms and prove the theorem.

The explicit numeric cutoffs use rational arithmetic and integer bit lengths. A test $j \geq \text{bitlength}(N)$ proves $2^j > N$ without allocating an enormous power. Six checkpoints verify both the joint stock and stopping condition; monotonicity beyond the supplied threshold follows from the retained real-log comparison. Native constants, prime existence, catalogue domination and strict asymptotic absorption retain additional eventual thresholds. None of these numbers is a practical running-time estimate.

7 Reproduction and limits

Run `python3 research/semantic-bulk/verify.py`. It checks all recorded PR21 theorem/code hashes and the pinned RaD source manifest, reconstructs the actual complex scalar charge and new stock, and produces `research/semantic-bulk/certificate.json`. The complete previous finite producers are reused by hash, without repeating their unchanged large computations. New routing, phase-inverse, bulk and semantic controls are retained separately under `research/semantic-bulk/controls`. Their finite checks discriminate incorrect endpoints, masking, rounding, period handling and row-stock reuse; the all-size claims also require the written proofs identified above.

The present declared finite bit saving bounds this composed witness below 2^{-16} . This does not exclude a better finite network or compiler. Future work at that threshold must improve the actual bit characteristic and recheck the complex leaf headroom; an optimistic role budget is not a multiplication bound.

References

- [1] Zhihao Chen. Translated partial-swap frames. <https://github.com/CrocSwap/integer-mult-bounds/pull/21>.
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- [3] Rohan Arun. Partial-swap ternary geometry. <https://github.com/CrocSwap/integer-mult-bounds/pull/19>.
- [4] R. C. Baker, G. Harman and J. Pintz. The Difference Between Consecutive Primes, II. Proc. London Math. Soc. 83 (2001), 532–562, Theorem 1. <https://www.cs.umd.edu/~gasarch/BLOGPAPERS/BakerHarmanPintz.pdf>.